Vinayak Tiwari

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Education

Kalinga Institute of Industrial Technology *2024-2028*
Bachelor of Technology in Computer Science Engineering | Bhubaneswar, Odisha

Experience

Chief Technology Officer (CTO) | InseLabs
- Led development of **AI-powered tools** and **workflow automation**, translating business needs into product features.
- Engineered features for **TrueHuman's 6-stage identity-protection workflow**: facial enrollment, embedding, web monitoring, matching, **0–100 risk scoring**, and takedown drafting.
- Architected **AI application workflows** and product integrations, guiding engineering decisions and technical delivery.

Academic Associate & Web Developer | International India Women's Programming Contest (IIWCPC)
- Authored and reviewed **algorithms and data structures** problems with the **IIT Kanpur team** for IIWCPC.
- Headed the **Contest Integrity Team**, overseeing competition fairness, security, and integrity.
- Developed the **official IIWCPC website** and coordinated contest preparation with technical and academic teams.

Technical Skills

Languages: Python, C++, C, JavaScript, TypeScript.
AI & LLMs: Azure OpenAI, Gemini API, LangChain, RAG, Prompt Engineering, Context Windows.
Retrieval & Data: Azure AI Search, Semantic Search, Vector Embeddings, Vector Databases, PostgreSQL, MongoDB.
Web & APIs: FastAPI, Pydantic, Zod, React, Next.js, Supabase, Tailwind CSS.
Cloud & Tools: Docker, Kubernetes, AWS, Azure, Vercel, Git, VS Code.
Concepts: Data Structures & Algorithms, Object-Oriented Programming, DevOps, Generative AI.

Projects

Weapon Detection | (Github) *Oct 2024 - Jan 2025*
Python , Open Cv , Haar Cascade Classifier
- Developed a real-time weapon detection system using **OpenCV** and **Haar Cascade classifiers** for live webcam and CCTV feeds.
- Trained a custom dataset of **255+ weapon images** to accurately detect pistols, rifles, and sniper weapons.
- Implemented anomaly-based surveillance detection to identify suspicious weapon activity in sensitive and crowded areas.

InvestIQ | (GitHub) *May 2025 - June 2025*
Python, FastAPI, Azure OpenAI, LangChain, PostgreSQL
- Built a **RAG-based financial intelligence platform** using Azure OpenAI and Azure AI Search for annual-report Q&A with company/year-filtered retrieval.
- Engineered PDF ingestion with **PyMuPDF4LLM, LangChain semantic chunking, and Azure OpenAI embeddings** to index financial reports in Azure AI Search.
- Automated KPI extraction with **Pydantic-validated LLM outputs** and PostgreSQL storage; containerized FastAPI services using Docker with Kubernetes configuration.

AlgoGym | (Live) *Aug 2024 - Sep 2024*
Next.js, TypeScript, Gemini API, Supabase, PostgreSQL
- Engineered a **Next.js 16, React 19, and TypeScript** platform with **8 algorithmic training tracks**, topic/rating filters, and verified Codeforces progress tracking.
- Integrated **Gemini LLM APIs** for learning paths, contextual tutoring, and code variants in **3 languages (C++, Java, Python)**, with **Zod-validated outputs** and PostgreSQL lesson caching.
- Built **Codeforces API and LeetCode GraphQL** synchronization with leased database jobs, retry handling, **Supabase Auth**, and PostgreSQL row-level security.

Achievements

- **ACM ICPC Amritapuri Asia West Regionals** qualifier (rank < 200) Certificate.
- **Codeforces Specialist (1430, top 8%)** — Vinayak Tiwari.
- **Codechef(1513, top 10%)** — Symmetry9.
- Secured **AIR 1** in Code O Olympic By **IIIT bhubneshwar.**
- Achieved **Rank 1** in Geek Clash Programming contest By **GFG-KIIT**
- Secured **Rank 1** among **100 teams** in **The Last Test Case**, an offline contest by **GFG-KIIT**.
- **Finalist** in Shaastra Programming Contest By **IIT Madras.**